SIT707 Software Quality and Testing

HD Task: Continuous integration and deployment (CI/CD) with Google cloud platform (GCP)

Overview

CI/CD in Google Cloud Platform (GCP) is a set of tools that automate the process of building, testing, and deploying software. CI/CD stands for continuous integration and continuous delivery or deployment. GCP supports software built with a range of programming languages including Python and Java.

In this task, you will configure GCP to implement CI and CD pipelines using your choice of application software (Python, Java or other application). The pipeline should demonstrate test failures, as well as pass all the test cases and finally build artefact and deploy. A code repository, such as GitHub, should be integrated to trigger the CI/CD process when a code change is pushed to the repository. You will prepare a well-formatted report describing your project steps and a video tutorial uploaded to YouTube to help others.

IMPORTANT: The configuration of GCP CI/CD involves challenging steps and permission issues. You are encouraged to assess them as early as possible and communicate with tutors/unit chair to have necessary access.

Instructions

For this task, you will need to

1. Explore GCP components required for CI/CD and validate if you have all the necessary permissions. In case there are additional GCP permissions required, which you should investigate well ahead of time, communicate with tutors/unit chair to have necessary access. Document any errors you encounter to include in your report.
2. Write an application of your choice and supported by the GCP for deployment. The application should contain a number of test cases.
3. You can think of using Google Cloud Build which is a service that builds, tests, and deploys code on Google Cloud.
4. Add tests with failures so your CI/CD pipeline generates errors, document those errors to include in your report.
5. The CI/CD pipeline should include a code repository where any update of your code triggers the CI/CD pipeline.
6. Now as you have already sorted out how to deal with GCP CI/CD, prepare a short 5–10-minute video tutorial helping others how to do the same and upload it to YouTube.

7. Optionally (if you are a hard worker), repeat the above steps for another type of application. For example, if you worked for Python application deployment, now try Java application or vice versa. You should include the 2nd application along with the first one in all the sections in your report divided into subsections. Provide contrastive discussion in sections where appropriate.

Submit a well formatted report as a pdf file combining below items -

- Chapters include executive summary, introduction, method, results and discussion. The introduction should describe at least GCP as a cloud platform, its modules with relating to CI/CD and GitHub integration. Method should include GCP configuration and other related configuration steps such as GitHub integration, Google cloud SDK installation and more.
- The Method section should include your application codes having test functions specifying functions which will fail and how to make it pass.
- The Results section should include all the events as evidence, screenshot of any output or error messages with proper caption and description.
- You should document every aspect of your work with enough justification to expect better outcomes.
- YouTube video tutorial URL should be included in your Results section and the Discussion section should include critical analysis of the overall process and any recommendations.

Submit your work

When you are ready, login to OnTrack and submit your pdf which consolidates all the items mentioned in the submission detail section above. Remember to save and backup your work.

Complete your work

After your submission, your OnTrack reviewer (tutor) will review your submission and give you feedback in about 5 business days. Your reviewer may further ask you some questions on the weekly topics and/or about your submissions. You are required to address your OnTrack reviewer's questions as a form of task discussions. Please frequently login to OnTrack for the task ***Discuss/Demonstrate*** or ***Resubmit*** equivalent to fix your work (if needed) based on the feedback to get your task signed as ***Complete***.